

SIMATS ENGINEERING
ASSESSMENT TOOL 1: Concept Mapping

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1522

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COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions. (BL2)*

Assessment Tool Weightage: 20%

Dave's Taxonomy: Imitation (Level 1)
Question Count: 4

Total Marks: 50

Code of Conduct

I _____ Hariprasath(192421246)_____ (Name / Reg. No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: _Hariprasath_____

Assessment Tasks

Concept Mapping Tasks

Create concept maps for the following tasks. Each map must show key nodes, link labels, and a logical hierarchy.

1. Concept Map 1: Cloud Computing Fundamentals. Include definition, characteristics, benefits, and limitations.
2. Concept Map 2: Evolution of Cloud Computing. Connect hardware evolution, internet software evolution, distributed systems, and service delivery.
3. Concept Map 3: Service Models. Map IaaS, PaaS, SaaS, and XaaS with provider responsibilities and user responsibilities.
4. Concept Map 4: Virtualization and Web Services as Enablers of Cloud. Show how these two concepts support cloud deployment and scalability.

Expected concept nodes:

- Elasticity, scalability, resource pooling, on-demand access, measured service
- Virtual machine, hypervisor, abstraction, API, SOAP, REST, provider, consumer
- Infrastructure layer, platform layer, application layer, shared responsibility

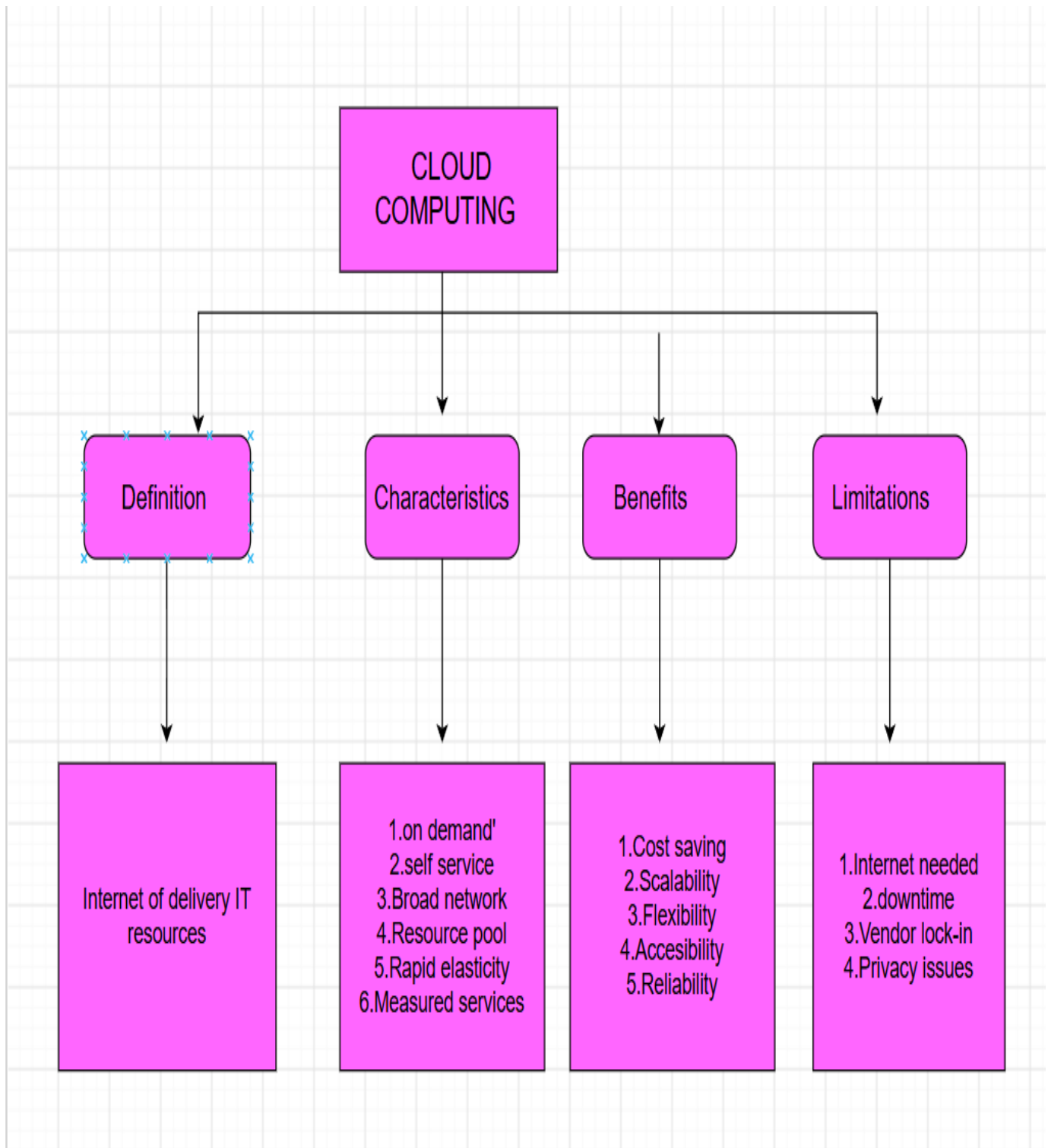
Concept Mapping Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Concept Identification	25%	All major concepts are accurately identified	Most concepts are identified correctly	Limited concepts are identified	Essential concepts are missing
Relationship Mapping	25%	Links between concepts are accurate and meaningful	Most links are correct	Some links are weak or unclear	Relationships are mostly incorrect
Hierarchy and Organization	20%	Excellent structure with clear hierarchy	Mostly organized with minor issues	Basic structure with weak hierarchy	No logical organization
Technical Relevance	20%	Map strongly reflects cloud course content	Mostly aligned to topic	Partially aligned to topic	Weak alignment to topic
Visual Clarity	10%	Neat, readable, and easy to interpret	Generally readable	Somewhat cluttered	Poorly presented

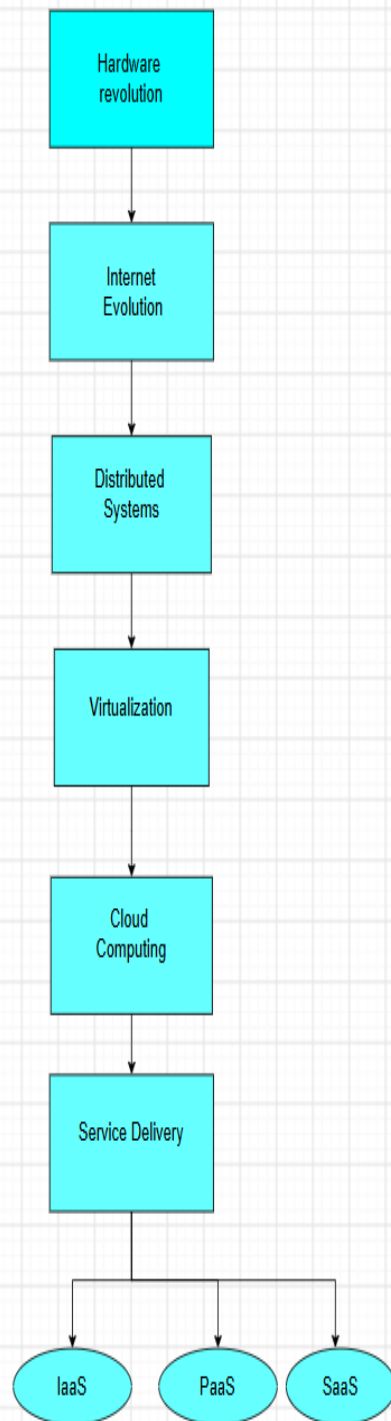
5. Questions

- i). Evaluate the effectiveness of your concept map in representing cloud computing principles. Which design decisions most strongly support your solution, and why?
- ii). Analyze how scalability, elasticity, and resource management influence the architecture shown in your concept map. What challenges could arise if these capabilities were absent?
- iii). Justify the selection of the APIs, web services, or cloud services included in your concept map. How do they contribute to the overall functionality and efficiency of the system?
- iv). Critically examine the relationships between the components in your concept map. How do these interactions improve reliability, performance, or user experience?
- v). Reflect on the process of developing your concept map. How has creating and analyzing the map changed your understanding of cloud computing architecture, and what improvements would you make if you were redesigning it?

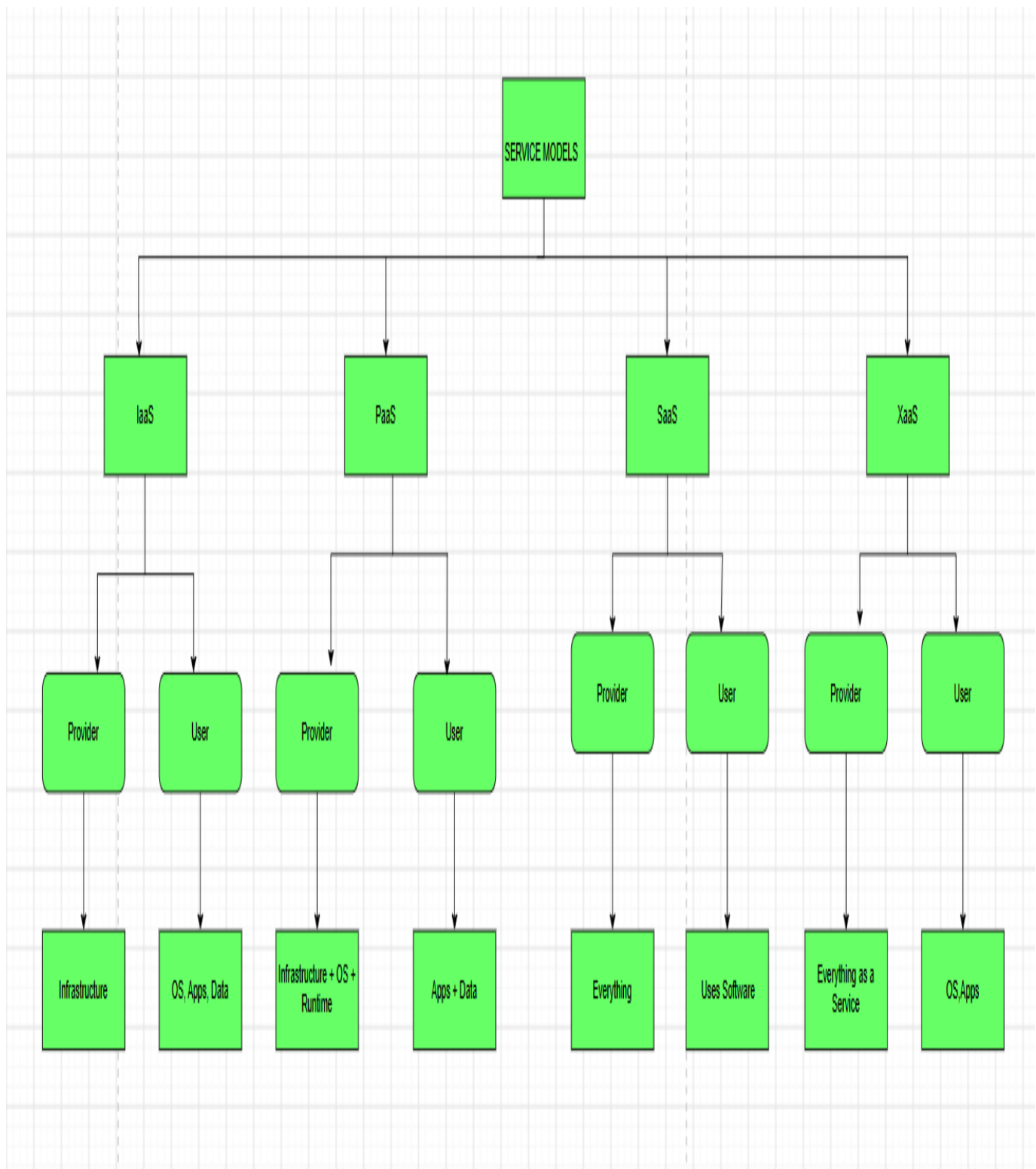
Concept Map 1: Cloud Computing Fundamentals. Include definition, characteristics, benefits, and limitations.



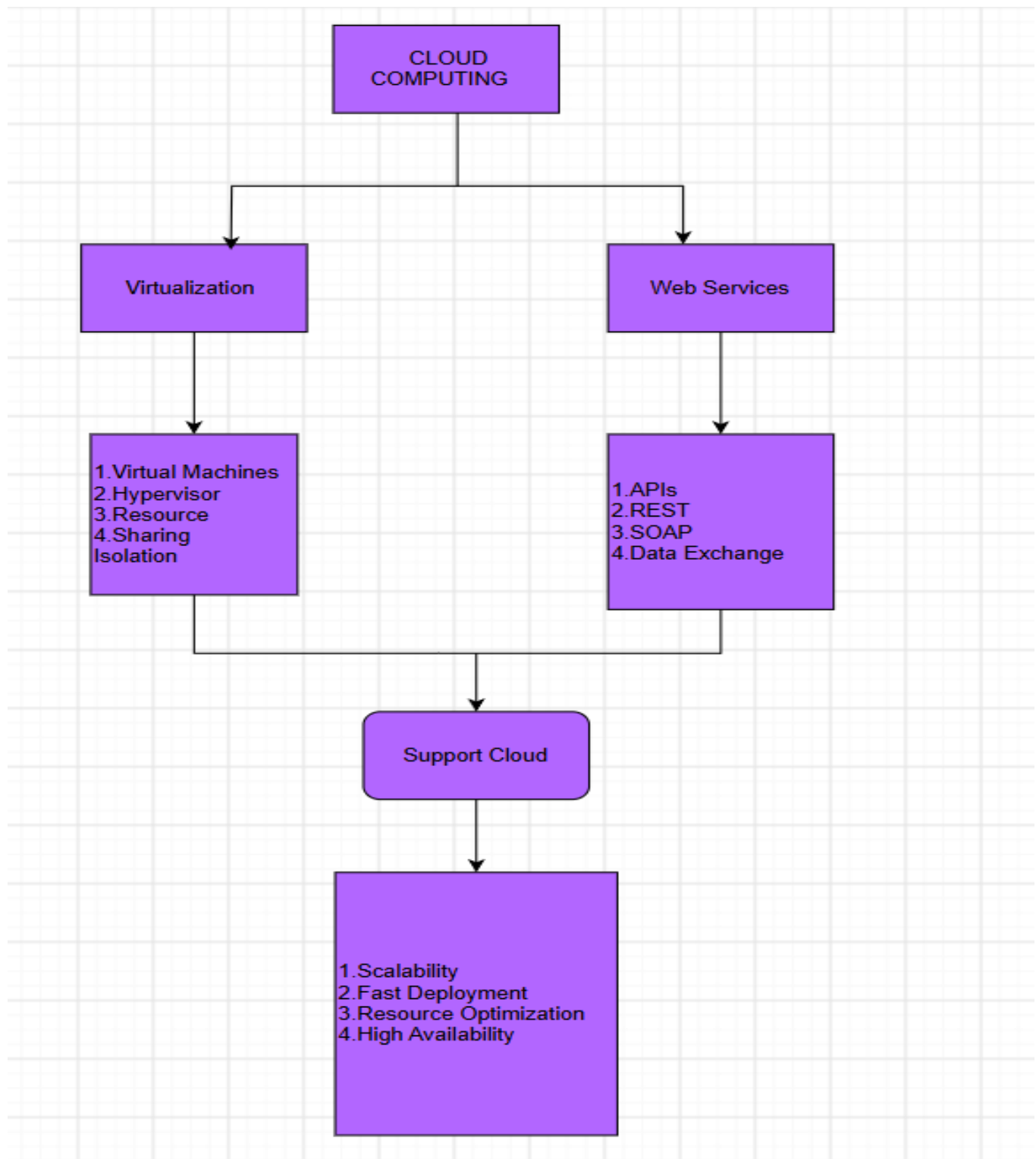
Concept Map 2: Evolution of Cloud Computing. Connect hardware evolution, internet software evolution, distributed systems, and service delivery.



Concept Map 3: Service Models. Map IaaS, PaaS, SaaS, and XaaS with provider responsibilities and user responsibilities



Concept Map 4: Virtualization and Web Services as Enablers of Cloud. Show how these two concepts support cloud deployment and scalability



5. Questions

i) Evaluate the effectiveness of your concept map in representing cloud computing principles. Which design decisions most strongly support your solution, and why?

The concept map effectively represents cloud computing by organizing the main concepts such as cloud fundamentals, service models, virtualization, and web services in a logical structure. The use of arrows shows the relationship between different components, making it easy to understand. Grouping related concepts together improves clarity and helps explain how cloud computing works. These design decisions make the concept map simple, organized, and easy to learn.

ii) Analyze how scalability, elasticity, and resource management influence the architecture shown in your concept map. What challenges could arise if these capabilities were absent?

Scalability allows the cloud to increase or decrease resources based on user demand. Elasticity automatically adjusts resources when workloads change. Resource management ensures efficient allocation of computing resources. Without these features, the system may experience slow performance, resource wastage, higher costs, service interruptions, and poor user experience during high traffic.

iii) Justify the selection of the APIs, web services, or cloud services included in your concept map. How do they contribute to the overall functionality and efficiency of the system?

APIs and web services enable communication between cloud applications and users. REST and SOAP web services support secure data exchange between different systems. Cloud services such as IaaS, PaaS, and SaaS provide infrastructure, development platforms, and software according to user needs. These services improve flexibility, automation, integration, and overall system efficiency while reducing development and maintenance costs.

iv) Critically examine the relationships between the components in your concept map. How do these interactions improve reliability, performance, or user experience?

The concept map shows that virtualization provides efficient resource utilization, while web services enable communication between applications. Service models deliver resources according to user requirements. Together, these components improve system reliability through redundancy, increase performance by efficient resource allocation, and enhance user experience by providing fast, scalable, and always-available services.

v) Reflect on the process of developing your concept map. How has creating and analyzing the map changed your understanding of cloud computing architecture, and what improvements would you make if you were redesigning it?

Creating the concept map helped me understand how different cloud computing components are connected and how they work together to provide cloud services. It improved my knowledge of service models, virtualization, scalability, and web services. If I redesigned the map, I would include cloud deployment models (Public, Private, Hybrid, Community), security features, and real-world cloud providers such as AWS, Microsoft Azure, and Google Cloud to make it more detailed and practical.